



Tzchen (Edison) Wu

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Education

Bachelor of Science in Electrical Engineering, Minor in Japanese

Dec 2027

University of Nevada, Las Vegas

Technical Skills

UC OASiS Foundations – Semiconductor Processing & IC Fabrication

Apr 2026

Programming: SystemVerilog, C, C++, Swift

Software & Tools: Quartus II, ModelSim, LTspice, MATLAB, Arduino, Git, Linux/Ubuntu, Docker, SolidWorks

Hardware & Test: FPGA, breadboarding, soldering, oscilloscopes, function generators, DMMs, 3D printing

Languages: English (Native), Mandarin (Native), Japanese (Conversational)

Projects

Car Turn Signal Controller – SystemVerilog, Quartus II, FPGA

Sep 2024

- Designed a **7-state Moore FSM** turn-signal controller for a 1965 Ford Thunderbird in SystemVerilog
- Simulated and verified FSM operation and output waveforms using ModelSim
- Implemented the design in Quartus II on a **DE2-115 FPGA** and verified operation on physical hardware

USB Power Supply & Charging Circuit

Apr 2026

- Designed and prototyped a regulated 5V USB power supply using an **LM7805**, validating it in LTspice before breadboard testing and perfboard soldering
- Verified voltage and current on both using a **Keithley 2110 digital multimeter**
- Implemented USB charging and verified operation by charging iPhones and other USB devices

Eddie Calendar – iOS Calendar Application

May 2026 – Present

- Developing and daily driving a native Swift iOS calendar integrating **3 event sources** – Google Calendar, Apple Calendar, and local events
- Built **4 custom calendar views** with filtering and customizable visibility
- Integrated external data from Apple WeatherKit and a news API to generate daily look-ahead notifications

Personal AI Location Server – Ubuntu, Docker, Dawarich, Tailscale, Ollama

Mar 2026

- Deployed a self-hosted, containerized server to ingest, store, and visualize iPhone GPS data
- Integrated **Ollama** for local natural-language querying and analysis of location history
- Configured a **Tailscale** VPN for secure access and troubleshoot networking and mobile-to-server integration

Work Experience

UGym Technology Corp. – CES Sales Representative

Jan 2023 & Jan 2024

- Represented UGym Technology at 2 CES exhibitions with real-time Mandarin–English interpretation
- Demonstrated products, explained features, answered questions, and spoke with prospective customers

Surglasses – Engineering Intern

Feb 2022 – Mar 2022

- Soldered wires and connectors, applied heat-shrink, and followed schematics to assemble wiring
- Revised the English-language user manual for an electronic product originally written in Mandarin

Point Robotics Medtech Inc. – Engineering Intern

Jun 2021

- Modeled parts in **SolidWorks**, prepared them for **3D printing**
- Translated and revised a company press release from Mandarin to English

Leadership

Founder & President, UNLV Run Club

Aug 2026 – Present

- Founded a UNLV RSO, distributed and created 75 promotional flyers to recruit members
- Organized weekly group runs with up to 5 participants and arranged peer support for run leadership

UNLV Marching Band – Trombone Section Leader

Aug 2024 – May 2026

- Coordinated 19 trombone members, including 9 freshmen, and split the section into 2 balanced groups
- Led sectionals, tracked attendance, conducted uniform checks, and communicated expectations to the section